CHAPTER 1

# YOU ARE HERE

The entire Earth is but a point, and the place of our own habitation but a minute corner of it.

—MARCUS AURELIUS, ROMAN EMPEROR,

*MEDITATIONS*, BOOK 4 (CA. 170)

As the astronomers unanimously teach, the circuit of the whole earth, which to us seems endless, compared with the greatness of the universe has the likeness of a mere tiny point.

—AMMIANUS MARCELLINUS (CA. 330–395),

THE LAST MAJOR ROMAN HISTORIAN,

IN *THE CHRONICLE OF EVENTS*

The spacecraft was a long way from home, beyond the orbit of the outermost planet and high above the ecliptic plane—which is an imaginary flat surface that we can think of as something like a racetrack in which the orbits of the planets are mainly confined. The ship was speeding away from the Sun at 40,000 miles per hour. But in early February of 1990, it was overtaken by an urgent message from Earth.

Obediently, it turned its cameras back toward the now-distant planets. Slewing its scan platform from one spot in the sky to another, it snapped 60 pictures and stored them in digital form on its tape recorder. Then, slowly, in March, April, and May, it radioed the data back to Earth. Each image was composed of 640,000